Mengting Ai

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Education

University of Illinois Urbana-Champaign (UIUC)
Ph.D. in Information Sciences — Aug 2024 – May 2028
Advisor: Prof. Jingrui He
Master of Computer Science — Aug 2022 – Dec 2023

Xi'an Jiaotong University (XJTU) — Sep 2018 – Jul 2022
B.E. in Computer Science

Publications

* denotes equal contribution.

• Peer-Reviewed Conference Publications

C1 **Does Accuracy Equal Evidence? Reasoning Faithfulness Under KV Cache Compression**
Mengting Ai, Jingrui He, Yue Guo
EMNLP 2026 Findings | KV Cache Compression | Reasoning Faithfulness | 29% acc. rate

C2 **Learning What Matters: End-to-End Adaptive Calibration Data Selection for LLM Pruning**
Mengting Ai, Sirui Chen, Xinrui He, Maggie Zarekarizi, Kommy Weldemariam, Jingrui He
EMNLP 2026 Findings | LLM Pruning | Calibration Data Selection | 29% acc. rate

C3 **NIRVANA: Structured Pruning Reimagined for Large Language Model Compression**
Mengting Ai, Tianxin Wei, Sirui Chen, Jingrui He
COLM 2026 | LLM Efficiency | Structured Pruning | 29% acc. rate

C4 **Proposing Better Rollouts for On-Policy Distillation**
Sirui Chen, Yunzhe Qi, Haobo Xu, Jiaru Zou, Mengting Ai, Zihao Li, Jingrui He
NeurIPS 2026 | On Policy Distillation | Reinforcement Learning | 25% acc. rate

C5 **Training Agent to Scale Inference-Time Reasoning**
Yunzhe Qi, Sirui Chen, Jiaru Zou, Yanjun Zhao, Mengting Ai, Jingrui He
NeurIPS 2026 | Test-time Scaling | Reinforcement Learning | 25% acc. rate

C6 **Agentic Reasoning for Large Language Models**
Tianxin Wei, Ting-Wei Li, Zhining Liu, Xuying Ning, Ze Yang, Jiaru Zou, Zhichen Zeng, Ruizhong Qiu, Xiao Lin, Dongqi Fu, Zihao Li, Mengting Ai, Duo Zhou, Wenxuan Bao, Yunzhe Li, Gaotang Li, Cheng Qian, Yu Wang, Xiangru Tang, Yin Xiao, Liri Fang, Hui Liu, Xianfeng Tang, Yuji Zhang, Chi Wang, Jiaxuan You, Heng Ji, Hanghang Tong, Jingrui He
TMLR | Survey

C7 **MedConceal: A Benchmark for Clinical Hidden-Concern Reasoning Under Partial Observability**
Yikun Han, Joey Chan, Jingyuan Chen, Mengting Ai, Simo Du, Yue Guo
COLM 2026 | Clinical AI | Benchmark | 29% acc. rate

C8 **Event Graph-Based Reliable Clinical Data Imputation**
Xinrui He, Mengting Ai, Junting Wang, Curtiss Cook, Jingrui He
KDD 2026 | Clinical Data | Graph ML | 18.5% acc. rate

C9 **AutoTool: Dynamic Tool Selection and Integration for Agentic Reasoning**
Jiaru Zou, Ling Yang, Yunzhe Qi, Sirui Chen, Mengting Ai, Ke Shen, Jingrui He, Mengdi Wang
ICML 2026 | Agentic AI | Tool Use | 26.6% acc. rate

C10 **Competing Visions of Ethical AI: A Case Study of OpenAI**
Melissa Wilfley*, Mengting Ai*, Madelyn Rose Sanfilippo
iConference 2026 | AI Ethics | Case Study | 24.5% acc. rate

C11 **MC-Search: Evaluating and Enhancing Multimodal Agentic Search with Structured Long Reasoning Chains**

Xuying Ning, Dongqi Fu, Tianxin Wei, **Mengting Ai**, Jiaru Zou, Ting-Wei Li, Jingrui He

ICLR 2026 | Multimodal Agents | Benchmark | 1.2% acc. rate | Oral

C12 **Influence-Preserving Proxies for Gradient-Based Data Selection in LLM Fine-Tuning**

Sirui Chen, Yunzhe Qi, **Mengting Ai**, Yifan Sun, Ruizhong Qiu, Jiaru Zou, Jingrui He

ICLR 2026 | Data Selection | LLM Fine-Tuning Efficiency | 28.0% acc. rate

C13 **ResMoE: Space-Efficient MoE Module Approximation via Wasserstein Barycenter and Residual Restoration**

Mengting Ai*, Tianxin Wei*, Yifan Chen*, Zhichen Zeng, Ritchie Zhao, Girish Varatkar, Bita Darvish Rouhani, Xianfeng Tang, Hanghang Tong, Jingrui He

KDD 2025 | LLM Effieicny | Mixture-of-Experts | 18.7% acc. rate

C14 **APEX^2: Adaptive and Extreme Summarization for Personalized Knowledge Graphs**

Zihao Li, Dongqi Fu, **Mengting Ai**, Jingrui He

KDD 2025 | Knowledge Graphs | Summarization | 18.7% acc. rate

C15 **PyG-SSL: A Graph Self-Supervised Learning Toolkit**

Lecheng Zheng, Baoyu Jing, Zihao Li, Zhichen Zeng, Tianxin Wei, **Mengting Ai**, Xinrui He, Lihui Liu, Dongqi Fu, Jiaxuan You, Hanghang Tong, Jingrui He

CIKM 2025 | Graph SSL | Toolkit | 43.2% acc. rate

• Workshop Papers

W1 **Faster, Lighter, Better: Compressed Earth Observation Foundation Models**

Mengting Ai, Alice L. Fan, Xinrui He, Xinyu He, Sirui Chen, Yikun Han, Maggie Zarekarizi, Kommy Weldemariam, Hanghang Tong, Jingrui He

NeurIPS 2026 Workshop TCCML | Earth Foundation Model | Efficient Fine-tuning | Structured Pruning

W2 **Harnessing Image Question Dependence for Better VLM Test-time Reinforcement Learning**

Xinrui He, Ting-Wei Li, Junting Wang, **Mengting Ai**, Xinyu He, Hanghang Tong, Jingrui He

NeurIPS 2026 Workshop VLM4RWD2026 | Test-time RL | VLM

W3 **AutoTool: Dynamic Tool Selection and Integration for Agentic Reasoning**

Jiaru Zou, Ling Yang, Yunzhe Qi, Sirui Chen, **Mengting Ai**, Ke Shen, Jingrui He, Mengdi Wang

ICCV 2025 Workshop MMRAgI | Agentic AI | Tool Use | Best Paper Award

W4 **TASP: Preserving Training Dynamics in Transformers via NTK-Aware Structured Pruning**

Mengting Ai, Tianxin Wei, Jingrui He

ICLR 2025 Workshop SLLM | Structured Pruning | Efficient Fine-Tuning

• Preprints & Manuscripts

P1 **Improved State Space Model**

Mengting Ai, Supriyo Chakraborty, Kartik Balasubramaniam, Anirban Das, Berkcan Kapusuzoglu, Dong Yu

Under Review | Linear Attention | Pre-training

P2 **Radiohead: Remote Sensing Earth Foundation Model Fusion with Mixture-of-Experts**

Mengting Ai, Lecheng Zheng, Zihao Li, Xinrui He, Maggie Zarekarizi, Prasanth Meiyappan, Kommy Weldemariam, Jingrui He

Under Review | Earth Observation FMs | Model Fusion

P3 **MLP Fusion: Towards Efficient Fine-Tuning of Dense and Mixture-of-Experts Language Models**

Mengting Ai*, Tianxin Wei*, Yifan Chen*, Jingrui He

Under Review | Mixture-of-Experts | Efficient Fine-Tuning

P4 **Evo-Memory: Benchmarking LLM Agent Test-Time Learning with Self-Evolving Memory**

Tianxin Wei, Noveen Sachdeva, Benjamin Coleman, Zhankui He, Yuanchen Bei, Xuying Ning, **Mengting Ai**, Yunzhe Li, Jingrui He, Ed H. Chi, Chi Wang, Shuo Chen, Fernando Pereira, Wang-Cheng Kang, Derek Zhiyuan Cheng

Under Review | Self-Evolving Memory | Benchmark

P5 **Heterogeneous Scientific Foundation Model Collaboration**
Zihao Li, Jiaru Zou, Feihao Fang, Xuying Ning, **Mengting Ai**, Tianxin Wei, Sirui Chen, Xiyuan Yang, Jingrui He
Under Review · Scientific FMs · Model Collaboration

P6 **Harnessing Generalist Agents for Contextualized Time Series**
Zihao Li, Kaifeng Jin, Yuanchen Bei, Jiaru Zou, Avaneesh Kumar, Xuying Ning, Yanjun Zhao, **Mengting Ai**, Baoyu Jing, Hanghang Tong, Jingrui He
Under Review · Time Series · Agents Harness

P7 **Code as Agent Harness**
Xuying Ning, Katherine Tieu, Dongqi Fu, Tianxin Wei, Zihao Li, Yuanchen Bei, Jiaru Zou, **Mengting Ai**, Zhining Liu, Ting-Wei Li, Lingjie Chen, Yanjun Zhao, Ke Yang, Bingxuan Li, Cheng Qian, Gaotang Li, Xiao Lin, Zhichen Zeng, Ruizhong Qiu, Sirui Chen, Yifan Sun, Xiyuan Yang, Ruida Wang, Rui Pan, Chenyuan Yang, Dylan Zhang, Liri Fang, Zikun Cui, Yang Cao, Pan Chen, Dorothy Sun, Ren Chen, Mahesh Srinivasan, Nipun Mathur, Yinglong Xia, Hong Li, Hong Yan, Pan Lu, Lingming Zhang, Tong Zhang, Hanghang Tong, Jingrui He
Under Review · Code Agents · Survey

Community Service

2026 ICLR; NeurIPS (Top Reviewer); ICML (Silver Reviewer); COLM; TMLR; AAAI
2025 ICLR; TKDD; iConference (Reviewer & Moderator); FM4RoboPlan@RSS
2024 AAAI

Research Experience

Capital One — Jun 2026 – Aug 2026
Applied Research Intern
- Conducted research on the pretraining of state-space models.

Institute of Software, Chinese Academy of Sciences — Oct 2021 – Jan 2022
NLP Algorithm Intern
- Built a cybersecurity question-answering system using DAPT-BERT.
- Crawled security-related Stack Overflow questions and constructed a semantic-similarity training dataset.
- Developed a DAPT-BERT binary classifier and achieved 92% accuracy with the DAPT100k model.

Baidu — Feb 2021 – Aug 2021
NLP Algorithm Research Intern
- Built a three-layer XGBoost text classifier with Word2Vec features, achieving an F1 score of approximately 93%.
- Expanded an employee skill-tag pool with 4-grams and TF–IDF, increasing tag coverage from 50% to 80%.
- Improved hard-skill labels through BERT-based encoding and iterative centroid refinement.

Professional Experience

UIUC Prairie Research Institute — Jul 2023 – Aug 2024
Software Engineer Intern
- Modeled time-series rainfall and streamflow data with PyTorch.
- Developed an interactive map that calculates and displays watershed areas from user-selected streamflow points.
- Implemented the system with HTML, CSS, JavaScript, Leaflet, Node.js, and the ArcGIS Pro API.

TE Connectivity — Jun 2022 – Mar 2023
Digital Intern
- Researched emerging AI technologies, including ChatGPT, and presented findings to the digital team.
- Developed RESTful data-processing interfaces with FastAPI for CRUD operations on cloud-hosted MySQL and MongoDB databases.
- Deployed the Mingdao low-code platform on AWS EC2 with Docker and Nginx load balancing and reverse proxying.
- Automated SQL generation for backend data access and manipulation, substantially reducing development costs.

Awards

NeurIPS 2026 Top Reviewer — **Oct 2026**

University Nominee (3/all), Apple Scholars in AI/ML PhD Fellowship — **Sep 2026**

ICML 2026 Silver Reviewer — **May 2026**

UIUC Fall 2025 Facilitating Learning Excellence Reward (IS 577 Data Mining TA) — **Apr 2026**

ICCV 2025 MMRAgI Workshop Outstanding Paper — **Oct 2025**

KDD Student Travel Award — **Jun 2025**

Outstanding Student (top 15%), XJTU — **Oct 2021**

Third-Class Scholarship (top 20%), XJTU — **Oct 2020 & 2021**

2nd Place, KPMG × Yiyou Youth Center "STEM is HER Future" (2/48) — **Aug 2021**

Invited Talk

ResearchTrend.AI: On Structured Pruning — **Nov 2025**